

AERO40007 – Mechanics

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One degree of freedom vibrations

(1.) A cylinder with the axis vertical floats in a liquid of density ρ . It is pushed down by a small amount and then released. Find the period of oscillation if the cylinder has weight W and cross-sectional area A .

$$F_{\text{floats}} = \rho V g = \rho A \Delta L g$$

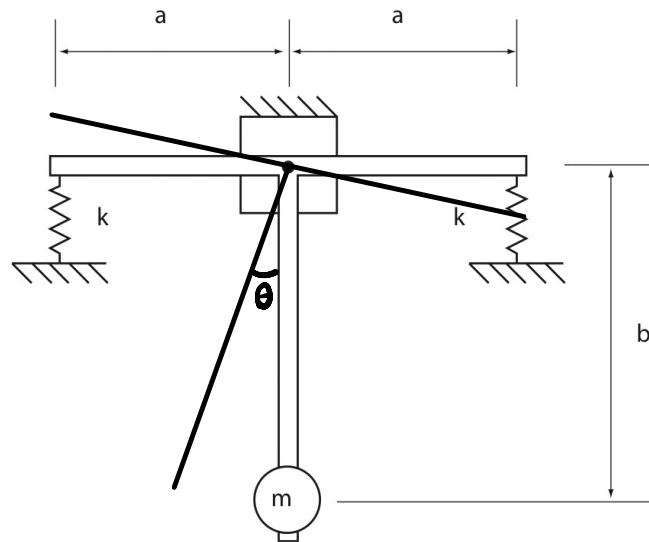
$$F = ma = m \Delta \ddot{L}$$

$$\therefore m \Delta \ddot{L} + \rho A g \Delta L = 0$$

$$\therefore \omega^2 = \frac{\rho A g}{m} = \frac{\rho A g^2}{W}$$

$$\therefore T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{W}{\rho A g^2}}$$

(2.) Derive the equation of motion for the rocker arm in the figure. The mass of the structure is negligible compared to the mass of the bob m . What is the natural frequency of the system?



$$H = I\omega = \int mv \, dt = Fd$$

$$\therefore I\omega = -mb^2\ddot{\theta}$$

$$Fd = ka \sin\theta \cdot a \times 2 + mg \sin\theta \cdot b = 2ka^2\theta + mgb\theta$$

$$\therefore mb^2\ddot{\theta} + (2ka^2 + mgb)\theta = 0$$

$$\therefore \omega_n^2 = \frac{2ka^2 + mgb}{mb^2}$$

$$\therefore f = \frac{1}{T} = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{2ka^2 + mgb}{mb^2}}$$

(3.) The free vibration of a spring mass system is to be changed such that:

(i) The period of vibration increases by at least 15%.

(ii) The amplitude of vibration decays by at least 95% after one cycle.

This change is produced by the addition of a dashpot in parallel with the spring. Find the damping ratio ζ of the new system.

$$\text{(i)} \quad \frac{T_{\text{damped}}}{T_{\text{undamped}}} = \frac{1}{\sqrt{1 - \zeta^2}} \geq (1 + 15\%) = 1.15$$

$$\therefore \sqrt{1 - \zeta^2} \leq \frac{20}{23} \quad \therefore 1 - \zeta^2 \leq \left(\frac{20}{23}\right)^2$$

$$\therefore \zeta^2 \geq 1 - \left(\frac{20}{23}\right)^2 \quad \therefore \zeta \geq \sqrt{1 - \left(\frac{20}{23}\right)^2}$$

$$\therefore \zeta \geq 0.4938$$

$$\text{(ii)} \quad \ln\left(\frac{x(t)}{x(t+T)}\right) = \frac{2\pi\zeta}{\sqrt{1 - \zeta^2}} \geq \ln\left(\frac{1}{1 - 95\%}\right) = \ln 20$$

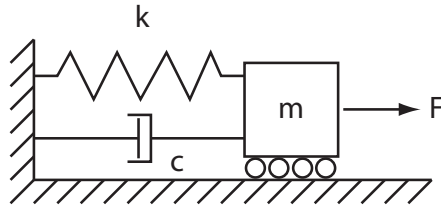
$$\therefore 4\pi^2\zeta^2 \geq (\ln 20)^2 - (\ln 20)^2\zeta^2$$

$$\therefore (4\pi^2 + (\ln 20)^2)\zeta^2 \geq (\ln 20)^2$$

$$\therefore \zeta \geq \sqrt{\frac{(\ln 20)^2}{4\pi^2 + (\ln 20)^2}}$$

$$\therefore \zeta \geq 0.4304$$

(4.) In the damped system shown in the figure a force $F = 8 \sin 2t$ acts on the block of mass $m = 30$ kg. The spring has stiffness $k = 600$ N/m, and the dashpot has damping constant $c = 75$ Ns/m. What is the amplitude of steady state motion and the maximum force transmitted to the wall?



$$m\ddot{x} + c\dot{x} + kx = F = 8 \sin 2t$$

$$\therefore F_o = 8 \text{ N} , \gamma = 2$$

$$\therefore A = \frac{F_o}{\sqrt{(k - m\gamma^2)^2 + (c\gamma)^2}}$$

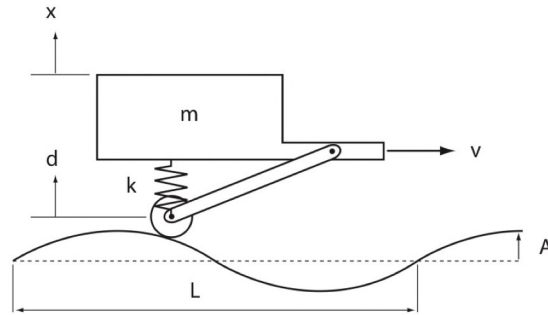
$$= 0.01591 \text{ m}$$

$$\therefore F_{\max} = \sqrt{(kA)^2 + (cVA)^2}$$

$$= A \sqrt{k^2 + (cV)^2}$$

$$= 9.8386 \text{ N}$$

(5.) A trailer of mass m runs along a corrugated road at constant speed v as shown. The road may be approximated as a sinusoid with amplitude A and wavelength L . The radius of the wheel is small, so the motion of the axle δ is closely approximated by the sinusoid. Assuming the trailer is fully supported by the linear spring, stiffness k , determine the steady state motion $x(t)$. What speed causes resonance?



$$\omega_n = \sqrt{\frac{k}{m}}$$

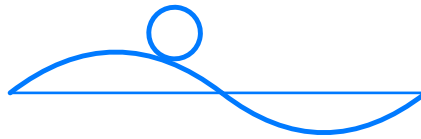
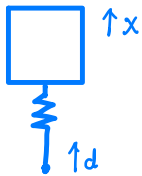
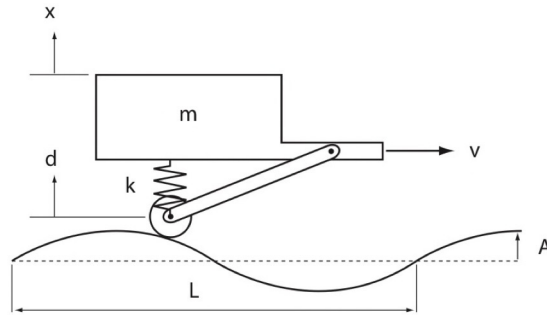
$\therefore$ For whole system,

$$(\omega_n)_t = \frac{L}{2\pi} \omega_n$$

$\therefore v \rightarrow \omega_n$ at resonance

$$\therefore v = \frac{L}{2\pi} \sqrt{\frac{k}{m}}$$

(5.) A trailer of mass m runs along a corrugated road at constant speed v as shown. The road may be approximated as a sinusoid with amplitude A and wavelength L . The radius of the wheel is small, so the motion of the axle δ is closely approximated by the sinusoid. Assuming the trailer is fully supported by the linear spring, stiffness k , determine the steady state motion $x(t)$. What speed causes resonance?



$$F_k = -k(x - d - \delta_0)$$

$$\begin{aligned} \therefore m\ddot{x} &= F_k + W \\ &= -k(x - d - \delta_0) - mg \end{aligned}$$

$$\therefore m\ddot{x} + kx - kd - k\delta_0 + mg = 0$$

$\therefore$ At equilibrium

$$-k\delta_0 + mg = 0$$

$$\therefore m\ddot{x} + kx = kd$$

$$T = \frac{L}{v}$$

$$\therefore \omega = \frac{2\pi}{T} = \frac{2\pi v}{L}$$

$$\therefore d(t) = A \sin\left(\frac{2\pi v}{L}t + \phi\right)$$

$$\therefore m\ddot{x} + kx = kA \sin\left(\frac{2\pi v}{L}t + \phi\right)$$

$\therefore v \rightarrow \omega_n$ at resonance

$$\therefore \frac{2\pi}{L}v = \sqrt{\frac{k}{m}}$$

$$\therefore v = \frac{L}{2\pi} \sqrt{\frac{k}{m}}$$

$$\therefore \begin{array}{c} \xrightarrow{Ak} \quad \xleftarrow{mv^2 x_0} \\ \xrightarrow{kx_0} \end{array}$$

($cvA = 0$ at x_0)

$$\therefore Ak + mv^2 x_0 = kx_0$$

$$\therefore x_0 = \frac{Ak}{k - mv^2}$$

$$\therefore v = \frac{2\pi}{L}v$$

$$\therefore x_0 = \frac{AkL^2}{kL^2 - 4\pi^2 mv^2}$$

$$\therefore x(t) = x_c(t) + x_p(t)$$

$$= A \sin\left(\frac{2\pi v}{L}t + \phi\right)$$

$$+ \frac{AkL^2}{kL^2 - 4\pi^2 mv^2}$$

Answers

1.

$$T = 2\pi\sqrt{\frac{W}{g^2\rho A}}$$

2.

$$\omega = \sqrt{\frac{mgb + 2ka^2}{mb^2}}$$

3.

$$\zeta = 0.494$$

4.

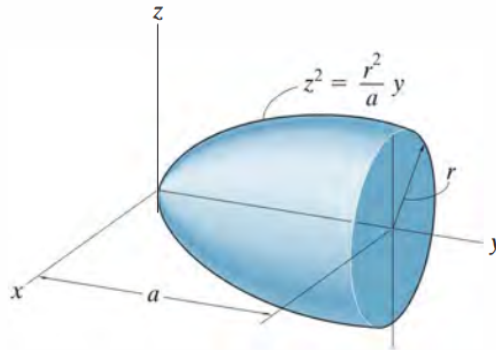
$$A = 1.591 \times 10^{-2} \text{ m}, \quad F_{max} = 9.84 \text{ N}$$

5.

$$v = \frac{L}{2\pi}\sqrt{\frac{k}{m}}$$

Moments of inertia

(6.) The total mass of the paraboloid of revolution is m . Calculate moment of inertia with respect to axis x and y .



Answer: $I_x = \frac{m}{6}(r^2 + 3a^2)$; $I_y = \frac{1}{3}mr^2$

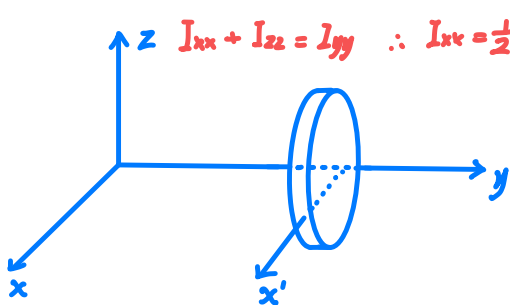
$$m = \int dm = \rho \int dV = \rho \int_0^a \pi z^2 dy = \pi \rho \int_0^a \frac{r^2}{a} y dy = \pi \rho \frac{r^2}{a} \int_0^a y dy$$

$$= \pi \rho \frac{r^2}{a} [\frac{1}{2}y^2]_0^a = \frac{1}{2}\pi \rho ar^2$$

$$I_y = \sum \frac{1}{2}mr^2 \text{ (for } \odot) = \frac{1}{2} \int_0^a (\rho \pi z^2) r^2 dy = \frac{1}{2} \int_0^a \rho \pi (\frac{r^2}{a} y) (\frac{r^2}{a} y) dy$$

$$= \frac{1}{2} \int_0^a \rho \pi \frac{r^4}{a^2} y^2 dy = \frac{\rho \pi r^4}{2a^2} \int_0^a y^2 dy$$

$$= \frac{\rho \pi r^4}{2a^2} \cdot \frac{1}{3}a^3 = \frac{1}{6}\rho \pi ar^4 = \frac{1}{3}(\frac{1}{2}\pi \rho ar^2)r^2 = \frac{1}{3}mr^2$$



$$I_x = I_{x'} + d^2m = \frac{1}{2}I_y + y^2m = \frac{1}{2} \cdot \frac{1}{3}mr^2 + y^2 \int dm$$

$$= \frac{1}{6}mr^2 + \int_0^a y^2 (\rho \pi z^2) dy = \frac{1}{6}mr^2 + \rho \pi \frac{r^2}{a} \int_0^a y^3 dy$$

$$= \frac{1}{6}mr^2 + \rho \pi \frac{r^2}{a} \cdot \frac{1}{4}a^4 = \frac{1}{6}mr^2 + \frac{1}{4}\rho \pi ar^2$$

$$= \frac{1}{6}mr^2 + \frac{1}{2}(\frac{1}{2}\rho \pi ar^2)a^2 = \frac{1}{6}mr^2 + \frac{1}{2}ma^2$$

$$= \frac{1}{6}(mr^2 + 3ma^2)$$